

Requirements to Subnet

- 1) Determine how many hosts you need.
- 2) Find the smallest $2^N \geq (\text{hosts} + 2)$
- 3) Host bits = $N \rightarrow \text{Prefix} = 32 - N$
- 4) Subnet size $\Rightarrow 2^N$
- 5) Usable Hosts = $2^N - 2$

Hosts Needed	Min 2^N	Prefix	Usable Hosts
2	4	/30	2
3-6	8	/29	6
7-14	16	/28	14
15-30	32	/27	30
31-62	64	/26	62
63-126	128	/25	126
127-254	256	/24	254

Subnet

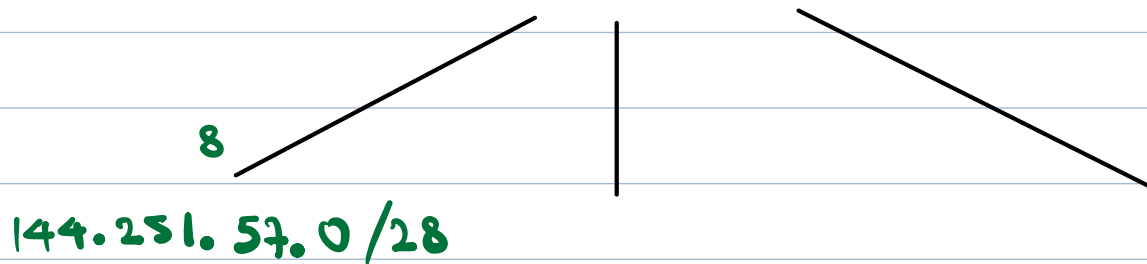
A smaller, logical piece of a larger computer network

Division

IP: 144.251.32.7 / 16

Network IP - 144.251.0.0

2^{16} - ~~X~~ unique hosts.



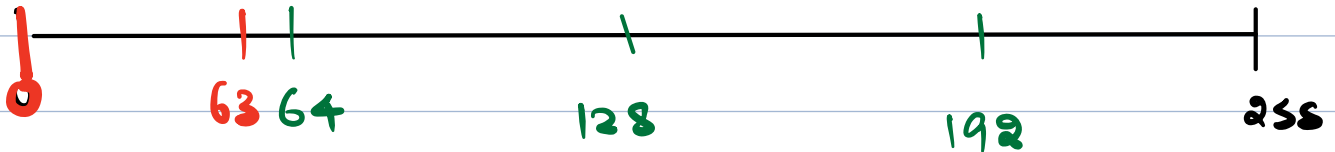
! Network IP : 192.168.1.0 / 24

3 subnets in this network with equal host division.

192.168.1.0 /24

$\frac{2^8}{2^2}$ 2^6

192.168.1.0/26



Subnet 1

192.168.1. 00 00 00 00 - 192.168.1. 0/26

192.168.1. 00 11 11 11 - 192.168.1. 63

Host $\Rightarrow$ 192.168.1.1 to 192.168.1.62

192.168.1. 01 00 00 00 $\rightarrow$ 192.168.1. 64/26

192.168.1. 01 11 11 11 $\rightarrow$ 192.168.1. 127

Host $\Rightarrow$ 192.168.1.65 to 192.168.1.126

192.168.1. 10 00 00 00 $\Rightarrow$ 192.168.1. 128/26

192.168.1. 10 11 11 11 $\Rightarrow$ 192.168.1. 191

Host $\Rightarrow$ 192.168.1.129 to 192.168.1.190

192.168.1. 11 00 00 00
192.168.1. 11 11 11 11

/24 $\Rightarrow$ 8 host bits $\Rightarrow 2^8$

/28 $\Rightarrow$ 4 $\Rightarrow 2^4$

$$\frac{2^8}{2^4} = 16 \text{ subnets}$$

VLSM $\Rightarrow$ Variable Length Subnet Masking.

VLSM allows different subnets to have different prefix lengths, enabling efficient use of IP spaces.

Golden Rule: Allocate the largest subnet first, then work downwards.

Network: 192.168.10.0/24

Dept	Hosts Needed	Prefix	Block size
Eng.	60	/26	64
Marketing	30	/27	32
HR	10	/28	16
Management	<u>5</u>	/29	8

1) Engineering

192.168.10.0 /26

Usable $\Rightarrow$.0 to .63

2) Marketing

192.168.10.64 /27

Usable $\Rightarrow$.64 to 95

3) HR

192.168.10.96 /28

Usable $\Rightarrow$.96 to .111

4) Management $\Rightarrow$ 192.168.10.112 /29

Usable $\Rightarrow$.112 to .119

IANA $\Rightarrow$ 2011 ran out of free blocks

IPv6 Addresses format

128 bit written as 8 groups of 4 hex digits separated by colons.

2001: 0db8: 85a3: 0000: 0000: 8a2e: 0370: 7334

$$(2^{32})^4 = 2^{128}$$

$\approx 3.4 \times 10^{38}$ addresses.

Compression Rules

$\Rightarrow$ Omit leading zeroes in each group

0db8 $\Rightarrow$ db8

0000 $\Rightarrow$ 0

2) Replace one seq. of consecutive all zero groups with ::

2001: 0db8: 85a3: 0000: 0000: 8a2e: 0370: 7334

2001: db8: 85a3:: 8a2e: 370: 7334

! :: can only be used once per address (otherwise ambiguous)

Common IPv6 Addresses

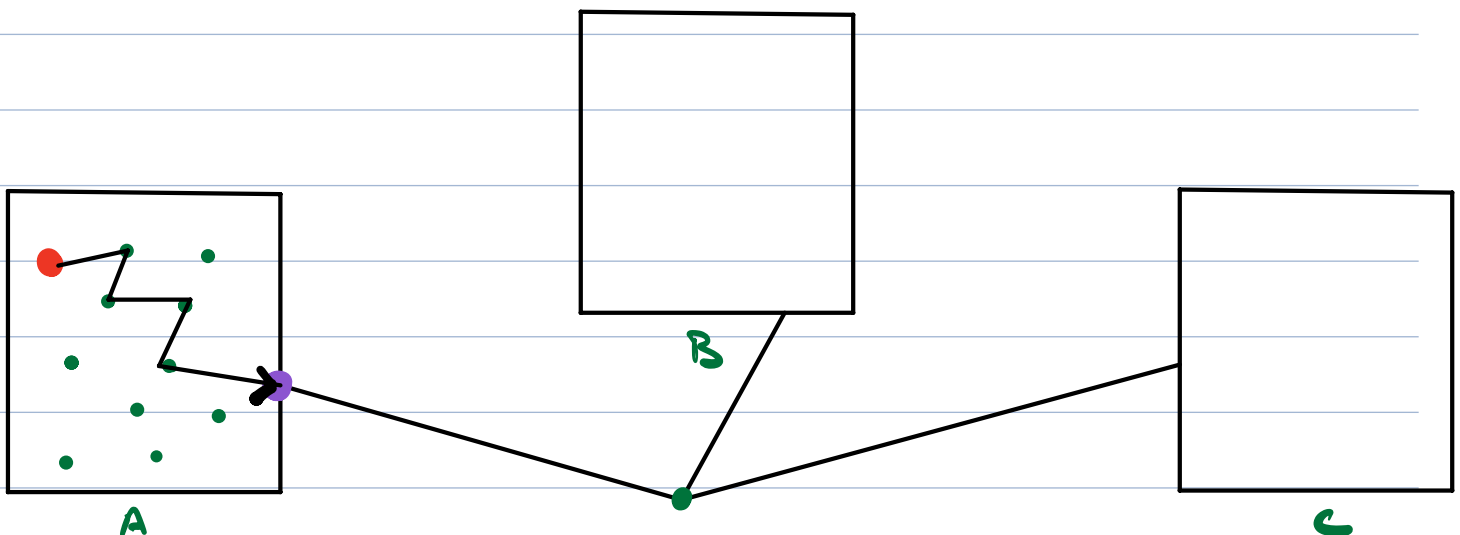
Common IPv6 Addresses

Address	Meaning
::1	Loopback (like 127.0.0.1)
::	Unspecified / any
fe80::/10	Link-local (auto-configured, not globally routed)
ff00::/8	Multicast
2000::/3	Global unicast (public internet)

IPv4 vs IPv6 Comparison

Feature	IPv4	IPv6
Size	32 bits	128 bits
Format	Dotted decimal	Colon-hex
Total addresses	~4.3 billion	3.4×10^{38}
Header	Variable (20–60 bytes)	Fixed 40 bytes
Broadcast	Yes	No (uses multicast)
NAT	Necessary	Not needed
ARP	Yes	Replaced by NDP
Fragmentation	Routers + hosts	Source host only

Default Gateway.



192.168.1.28 /24



192.168.1.126 /24

192.168.1.28 /24



192.168.31.126 /24

Linux

ip route show

Output: "default via 192.168.1.1 dev eth0"

Windows

route print

or ipconfig → "Default Gateway"

macOS

netstat -rn | grep default

NAT $\Rightarrow$ Network Address Translation